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| <br><b>UNIVERSITY OF TECHNOLOGY - VNUHCM</b><br><b>FACULTY OF COMPUTER SCIENCE AND</b><br><b>ENGINEERING</b> | <b>HOMEWORK 3</b> |                  | Semester/Academic year |  | 1          | 2026-2027 |
|                                                                                                                                                                                               |                   |                  | Due                    |  | 07/10/2026 |           |
|                                                                                                                                                                                               | Course title      | Machine Learning |                        |  |            |           |
|                                                                                                                                                                                               | Course ID         | CO3117           |                        |  |            |           |
|                                                                                                                                                                                               | Duration          | Take-home        | Homework set           |  | HW3        |           |
| <b>Notes:</b> Handwritten homework, submitted as homework/hw3-submission.pdf with tag hw3.                                                                                                    |                   |                  |                        |  |            |           |

### Instructions

- **Artificial neural networks** (Chapter 3). Five problems, each worth 2 points (0.5 per part); total 10 points.
- **Write by hand**, on paper or with a stylus. Show your working: a bare number earns no credit. Put your full name, student ID and the set number at the top of the first page.
- **Work on your own.** Do not use AI tools, solution manuals or classmates' work before your submission is tagged. You may use your lecture notes and the course slides. Any later AI use goes in AI\_USE.md, as for the weekly drills.
- **Submit in your course repository.** Scan all pages, in order, into one PDF of at most 5 MB, named homework/hw3-submission.pdf. Commit it, run python tools/check.py --checkpoint hw3, then git tag hw3 and git push origin main hw3.
- **Deadline: 07/10/2026 07:00** (the time GitHub receives the tag hw3). Solutions are released on 07/10/2026. Afterwards you may add homework/hw3-corrections.md; never replace or re-tag the submission.

**Problem 1 (L.O.2) (2 pts):** Answer the following questions about the perceptron. Use the convention  $x_0=1$  for the bias input and outputs in  $\{-1, +1\}$ .

- Write the output function of a perceptron and the perceptron training rule.
- The current weights are  $(w_0, w_1, w_2) = (0.2, -0.4, 0.1)$  and the learning rate is  $\eta = 0.1$ . For the training example  $(x_1, x_2) = (1, 1)$  with target  $t = +1$ , compute the output and the updated weights.
- Can a single perceptron represent the Boolean function XOR? Justify your answer.
- How does the delta rule differ from the perceptron training rule, and when does each one converge?

**Problem 2 (L.O.2) (2 pts):** Answer the following questions about activation functions in neural networks:

- Write the sigmoid, tanh and ReLU functions and their derivatives.
- The largest value of the sigmoid derivative is 0.25. Explain what this implies for backpropagation through 10 sigmoid layers, and give an upper bound on the product of the 10 derivative factors.
- Why does ReLU reduce this problem? What is the “dying ReLU” problem, and how does Leaky ReLU address it?
- For the pre-activations  $z = -2$ ,  $z = 0.5$  and  $z = 3$ , compute the outputs of ReLU, Leaky ReLU with slope 0.01, and the sigmoid.

**Problem 3 (L.O.2) (2 pts):** Answer the following questions about backpropagation. A network has inputs  $x_1=1$  and  $x_2=0.5$ , one hidden sigmoid unit  $h$  with weights  $w_1=0.5$  and  $w_2=-0.4$ , and one sigmoid output unit  $o$  with weight  $v=0.8$  from  $h$  (all biases are 0). The target is  $t=1$ , the loss is  $E = \frac{1}{2}(t - o)^2$ , and the learning rate is  $\eta = 0.5$ .

- Perform the forward pass: compute the output of the hidden unit and of the output unit.
- Compute the loss  $E$ .
- Compute the error term  $\delta_o$  of the output unit and the updated weight  $v$ .
- Compute the error term  $\delta_h$  of the hidden unit and the updated weight  $w_1$ .

**Problem 4 (L.O.2) (2 pts):** Answer the following questions about optimisation algorithms for training neural networks. Consider a single parameter  $\theta$  whose gradient is constant,  $g=2$ .

- A. Write the update rule of stochastic gradient descent and explain the role of the learning rate. What happens if it is too large or too small?
- B. With momentum  $v_t = \mu v_{t-1} + g_t$  and  $\theta \leftarrow \theta - \eta v_t$ ,  $\mu=0.9$ ,  $\eta=0.1$ ,  $v_0=0$  and  $\theta_0=1$ , compute  $v$  and  $\theta$  after two steps.
- C. For Adam with  $\beta_1=0.9$  and  $\beta_2=0.999$ , compute  $m_1$ ,  $v_1$  and the bias-corrected  $\hat{m}_1$ ,  $\hat{v}_1$  at the first step. What is the size of the first update in terms of  $\eta$ ?
- D. With plain SGD ( $\eta=0.1$ ,  $\theta_0=1$ , constant gradient  $g=2$ ), compute  $\theta$  after two steps and compare with part B. What does the comparison show about momentum?

**Problem 5 (L.O.2) (2 pts):** Answer the following questions about logistic regression and softmax regression:

- A. A logistic regression model has  $w_0 = -1$ ,  $w_1 = 2$ ,  $w_2 = -0.5$ . Compute  $P(y=1|x)$  for  $x = (x_1, x_2) = (1.5, 2)$  and the predicted class with threshold 0.5.
- B. If the decision threshold is raised from 0.5 to 0.8, what is the new decision boundary? How is it related to the boundary for threshold 0.5?
- C. Interpret  $w_1 = 2$  in terms of the odds  $P(y=1|x)/P(y=0|x)$ .
- D. A three-class softmax model outputs scores  $(2, 1, 0)$ . Compute the softmax probabilities and the cross-entropy loss (natural logarithm) if the true class is the second one.

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